

Prathamesh Balasaheb Kadam
Department of Artificial Intelligence and Data Science
AISSMS Institute of Information Technology
Pune, India

May 19, 2026

The Editor-in-Chief
Quantum Machine Intelligence
Springer Nature

Dear Editor,

We are pleased to submit our manuscript, “*Bell-Decomposition Quantum Kernels: A Multi-Scale Benchmark on Earth Observation and Beyond*,” for consideration as an original research article in *Quantum Machine Intelligence*.

Authors. Prajwal S. Gaikwad (ORCID 0000-0003-0805-9130); Prathamesh Balasaheb Kadam, corresponding author (ORCID 0009-0006-9211-4647); Shreyas Subhash Raut (ORCID 0009-0007-8900-9560); Tejas Uttam Shitole (ORCID 0009-0004-2175-8723). All four are with the Department of Artificial Intelligence and Data Science, AISSMS Institute of Information Technology, Pune, India. Per-author CRediT-style contributions are listed in the manuscript’s *Statements and Declarations*.

Contribution. The paper introduces the Bell-decomposition family of two-qubit quantum-kernel feature maps, unifying the singlet-only Self-Regulating Quantum Feature Map (SRQFM) and the four-Bell-sector Bell-Spectrum Coupling Map (BSCM) under an adjustable prior. The family sits inside the canonical two-body Pauli-rotation feature-map framework of Suzuki *et al.* (QMI 2020); within that framework, we prove two closed-form structural identities specific to the Bell-projector construction — a tight prior-independent bound $|\alpha| \leq 1/2$ on every per-pair Pauli coefficient, and the identity $\alpha_{ZZ} \equiv 0$ for the uniform prior (Theorem 1, full proofs in the Appendix, numerically unit-tested to 10^{-8}). The proof technique is elementary substitution of the standard two-qubit Bell–Pauli identities into the construction; the contribution is the construction itself — a usable closed-form quantum-kernel design device — and the design-relevant identities it admits.

Empirical results. At 16 qubits and $N = 10,000$ on So2Sat LCZ42 (17 classes) and EuroSAT (10 classes), the Bell-decomposition projected kernels add +0.167–+0.170 macro-F1 over the canonical commuting ZZ-PQK on So2Sat and +0.097–+0.108 on EuroSAT (Holm-corrected Wilcoxon $p = 0.0078$; complementary exact McNemar tests give $p \ll 10^{-100}$ on both datasets). The same architectural gap holds with positive sign on every split across four UCI tabular datasets (largest gap +0.646 accuracy on the 26-class Letter Recognition task), on a transverse-field Ising-model phase-classification benchmark (E49), on learned AE-16 deep features (E42), and under density-matrix simulation with realistic depolarising noise at $n = 8$ (E48, gap survives at +0.133–+0.136, Wilcoxon $p = 6.1 \times 10^{-5}$). Two complementary measurements (a 14-qubit BSCM build and a 16-qubit depth scan) reproduce the Thanasilp *et al.* concentration signature on real multi-class data.

Honest negatives. We make no claim of quantum advantage: at every scale tested, tuned classical RBF-SVM ties or exceeds the best Bell-decomposition kernel, consistent with recent

theory on bandwidth-tuning convergence (Canatar et al.; Flórez-Ablan, Roth and Schnabel) and prior empirical benchmarks (Bowles et al.; Schnabel and Roth, QMI 2025). The within-family architectural gap is reported as the main contribution: it is a design lesson for quantum-kernel architects — single-axis commuting ZZ generators carry a structural pathology that the broader non-commuting Pauli generator family does not — and is independent of whether quantum-vs-classical advantage is reached on a given task.

Fit with QMI. The paper is directly thematically aligned with the journal: it places its theoretical contribution within Suzuki *et al.*'s two-body Pauli-rotation feature-map framework (QMI 2020), it extends the benchmarking framework of Schnabel & Roth (QMI 2025) to a larger fixed-protocol multi-class Earth-observation setting, and it directly engages with the bandwidth-tuning convergence picture established by Flórez-Ablan, Roth and Schnabel (Quantum Sci. Tech. 2025). The combination of $n = 16$ qubits, $N = 10,000$, $K \in \{10, 17\}$ classes, and Holm-corrected paired significance tests on real remote-sensing data is, to our knowledge, the largest such benchmark in the QML kernel literature to date.

Reproducibility and open access. All experiment scripts, computed Gram matrices, JSON metric caches, the LaTeX source, and the unit-test suite (numerically validating every cited identity to 10^{-8}) are released at https://github.com/prathameshkadam130404/Quantum_Kernel.

Originality and exclusivity. We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission to *Quantum Machine Intelligence*, including the names and order of authors. No external funding was received. The authors declare no competing financial or non-financial interests.

We thank you for your consideration and look forward to the editorial decision.

Yours sincerely,

Prathamesh Balasaheb Kadam
(corresponding author)
Department of Artificial Intelligence
and Data Science
AISSMS Institute of Information
Technology, Pune, India
prathameshkadam130404@gmail.com